I. Describing Motion

A. Observing—You will see several demonstrations of motion of objects.
What sorts of things have to be specified to reproduce the motion acted out? Make a list of variables.

Some examples: starting position, final position, direction of movement, how far you travelled, how quickly you travelled, how much time it took.

B. Technical Terms—Quantities and qualities in physics tend to have clear and specific meanings, often related to measurement. Let's assign a technical vocabulary to the variables we identified above:

Initial position - x_i - where you started the motion
Final position - x_f - where you ended the motion
Speed - the rate at which the distance was covered
distance - how far you travelled
time interval - how much time it took

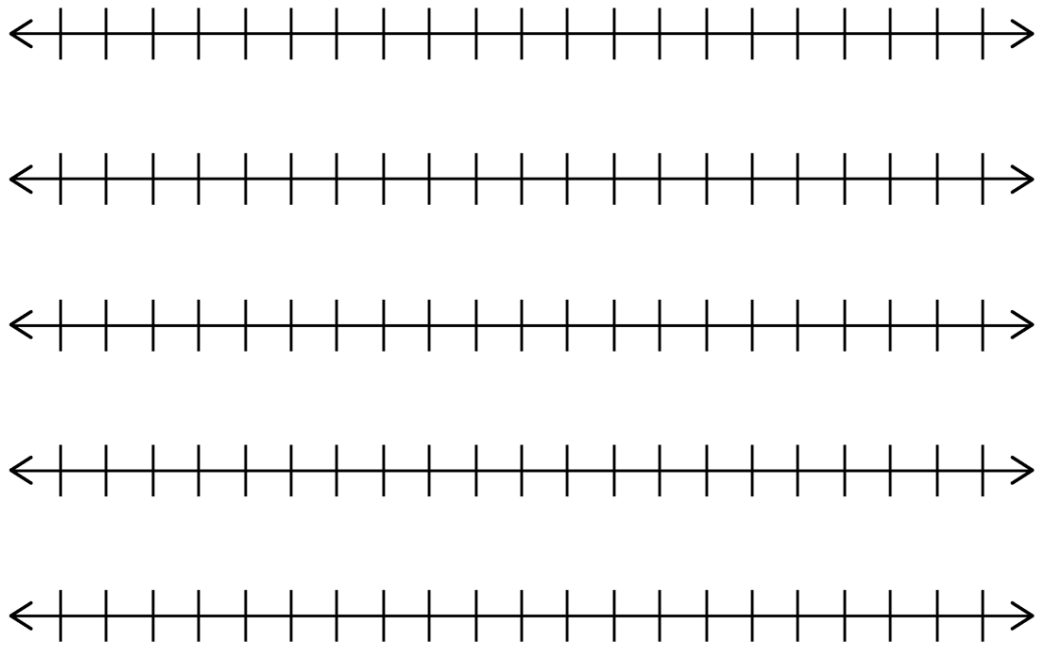
The word "rate" has a special meaning here. Whenever you see the word "rate" in a definition in this class, read it as "the ratio of some quantity with respect to time." This means that all rate quantities in this class will be fractions with time in the denominator.

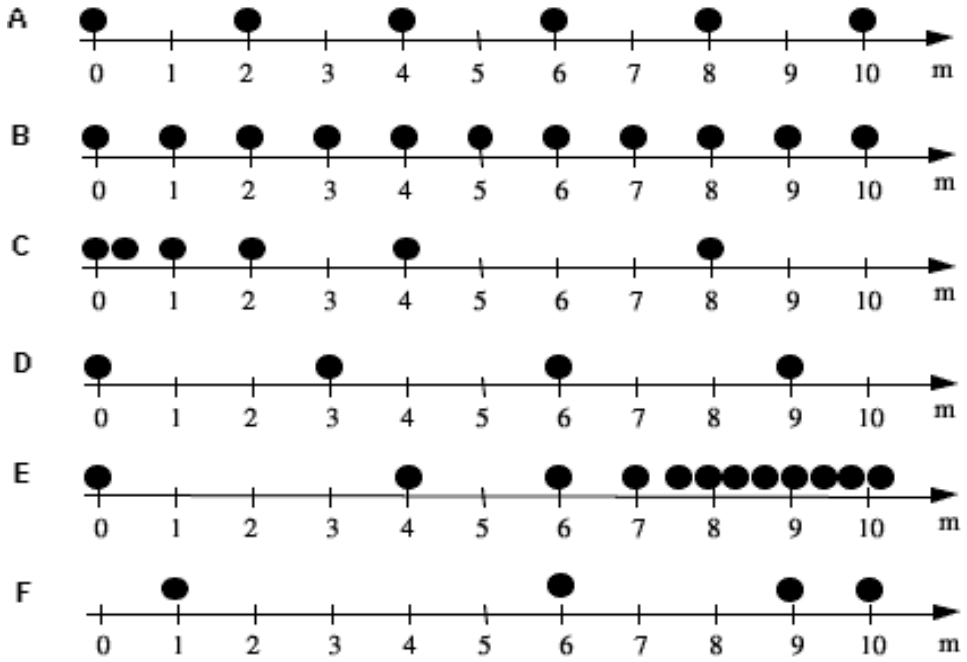
C. The Equation—Let's derive the equation for speed from its definition: speed is the **rate** at which a distance is covered.

$$s = \frac{d}{t}$$

s = speed (in m/s)
d = distance (in m)
t = time (in seconds)

D. Motion Diagrams—We will now learn a technique for visualizing motion. Imagine a strobe photograph is taken every second of a series of spheres moving from left to right. Your diagram will show the location of the sphere when each photograph is taken. Label the x axis in meters, and plot the motion of the sphere. We did a number of examples in class; check with a classmate or me for details.





Example 1: Take a look at the 6 motion diagrams above. Rank the 6 spheres (A - F) according to their displacement at 3 seconds:

Greatest _F*____
_D*____
_E____
_A____
_B____
Least _C____

Which sphere is going slowest in the beginning? ___C__
Which is going fastest? ___F__

Which sphere is going slowest at the **end**? ___E__
Which is going fastest? ___C__

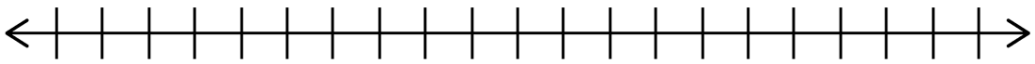
*F & D both have a displacement of 9 m at 3 seconds.

II. Adding direction

A. I start at East Brunswick High School and travel 100 miles. Where do I end up?
Anywhere on a 100 mile radius from EB - possibly in the ocean!

B. II. Setting up directions

a. Setting the center of the number line as zero, label numbers -5 to +5



b. From +1 on your number line, move -2, then +3, and finally -4. Where do you end up? __-2___

c. Does your answer to **b** depend upon your choice of positive? _Nope__

d. Convention: We will use positive (+) to represent motion to the right, and negative (-) for left.

III. But what is a negative?

a. Name some quantities that can be negative.
Position, temperature, velocity, displacement

b. Definitions of new terms

Vector quantities: those that have magnitude and direction (the direction can be + or -)
Scalar quantities: those that have ONLY magnitude (there is no direction involved)

IV. New quantities

a. In the Do Now, we lacked some information about which direction I was traveling for a particular distance. When you include information about the direction you traveled a particular distance, the resulting *vector quantity* is called **displacement**. Mathematically we can calculate this by subtracting our *starting position* from our *ending position*. Solve the three examples below. $\Delta x = x_f - x_i$; see definition at bottom of page*

Ex 1. An ant walks on a number line. It begins at the 1 cm mark and walks to the 5 cm mark. What is the ant's displacement? What is the distance the ant traveled?
Δx = 5 cm - 1 cm = +4 cm (4 cm **to the right**); distance = 4 cm

Ex. 2. An ant walks on a number line. It begins at the 6 cm mark and walks to the 4 cm mark. What is the ant's displacement? What is the distance the ant traveled?
Δx = 4 cm - 6 cm = -2 cm (2 cm **to the left**); distance = 2 cm

Ex. 3. An ant walks on a number line. It begins at the 4 cm mark and walks to the 6 cm mark. Finally, it walks to the 5 cm mark. What is the ant's displacement? What is the distance the ant traveled?
Δx = 5 cm - 4 cm = +1 cm (1 cm **to the right of x_i**); distance = 3 cm (this is a corrected answer)

b. Similarly, it is often helpful to include information regarding direction when discussing speed. Describe a situation where someone travels at a speed of 30 mph and never reaches their destination 60 miles away. What if the destination is to the east of you, and you got 30 mph to the North?

c. When we include direction with our definition of speed, we call this "speed" a *velocity*. It is defined as the **rate** at which a displacement occurs.

$$velocity = \frac{displacement}{time} = \frac{\Delta x}{t}$$

Where units are: velocity (m/s)
displacement (m); time (s)

Ex. 4. Look back to example 1. If the ant took 2 seconds to complete its journey, what was the ant's *speed?* The ant's *velocity?* How do these two numbers compare? Convert your units for speed and velocity into m/s.
speed = 2 cm/s = 0.02 m/s; velocity = 2 cm/s **to the right** = 0.02 m/s **to the right**; EVEN IF THE NUMBERS ARE THE SAME (as in this example), these are different measurements; speed is scalar, and velocity is a vector quantity.

Ex. 5. Look back to example 3. If the ant took 10 seconds to complete its journey, what was the ant's *speed?* The ant's *velocity?* How do these two numbers compare?
speed = 0.3 cm/s (corrected answer); velocity = 0.1 cm/s **to the right**.

V. Closure: We often use the terms distance, displacement, position, speed, and velocity to talk about real life situations. Come up with a real life situation where you would need to tell someone about each of the above quantities.

* Displacement is the change in position of an object; Δx = change in position = displacement

Average and Instantaneous quantities

I. Do Now: You and a friend are both driving in separate cars from your houses to the shore. You leave and arrive at the same time. Were you both traveling the same velocity? Why or why not? What assumptions did you make?
The only way you could have had the same velocity was if you underwent the same displacement; for that to happen, you would have had to leave from the same starting position and arrived at the same final position in the same amount of time. Since this is unlikely, you likely had different velocities. Your initial assumptions will vary.

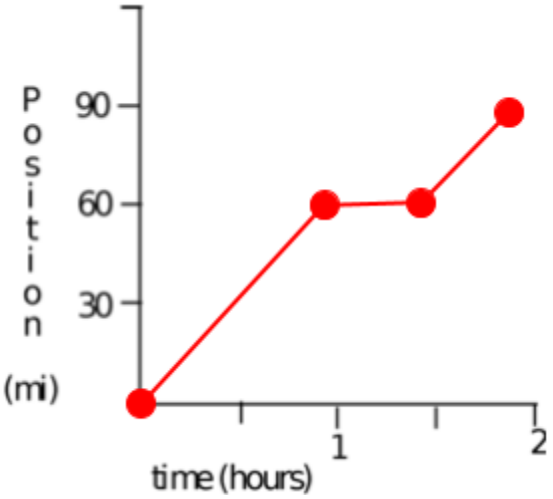
II. Translating into graphs and definitions

a. You travel between two cities that are 90 miles apart. You drive at 60 miles per hour for 1 hour, take a half-hour rest, then drive for another half-hour at the same speed (60 mph).

Your friend calculates your velocity and claims that it is 45 mph.

1) Explain why your friend is correct. $\frac{displacement}{time} = \frac{90\ miles}{2\ hours} = 45\ mph$

2) Explain why your friend is incorrect. He was never going at 45 mph during the trip (we are assuming an unrealistic situation here; that speed can change *instantly*).
3) Draw a position versus time graph to show what happened on your trip.



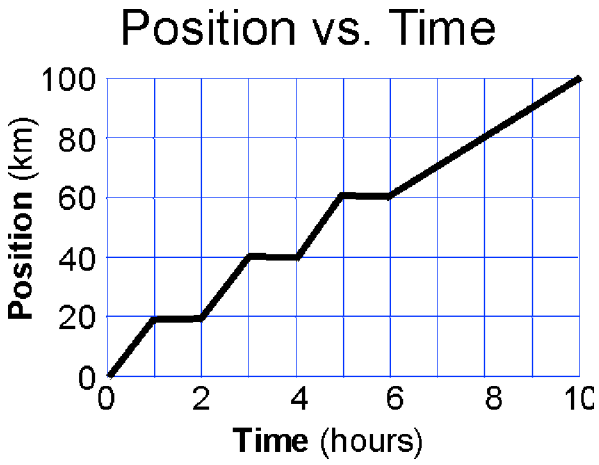
Average velocity: **the total displacement divided by the total time**

Instantaneous velocity: **the average velocity for a *very* small time interval (approaching zero); think of it as the velocity *at a given instant*.**

b. Using the mathematical definition for instantaneous velocity, describe how you would determine the instantaneous velocity from just a graph of position versus time. By finding the slope of a **very small** time interval.

III. Interpret a position vs. time graph

The position versus time graph at right shows a boat traveling through a long canal. The boat has to stop at locks for changes in water level.



a) How many stops does the boat make? Explain.

Three. There are three plateaus during which position remains the same.

Use this graph for the two questions on the next page.

b) What is the boat's average velocity for the whole trip? SHOW ALL MATH.

$\frac{100\ km - 0k\ m}{10h - 0h} = 10\ km/h$

c) What is the highest velocity the boat reaches? SHOW ALL MATH.

$\frac{40 - 20\ km}{3-2h} = 20\ km/h$; there is similar math for the first, third, and fifth segments, all of which give 20 km/h

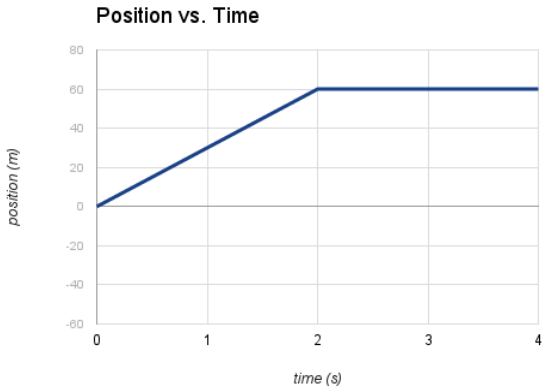
IV. Drawing velocity versus time from position versus time graphs

a. To find the velocity from a position versus time graph simply find the _slope___.

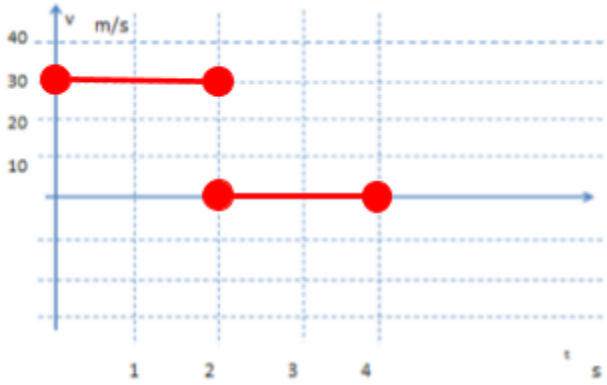
b. Find the slope_ of each section if there are multiple sections on the graph.

c. If the position versus time graph is a straight line, that means the velocity is _constant_ and should look like a _horizontal straight line_ on the velocity versus time graph.

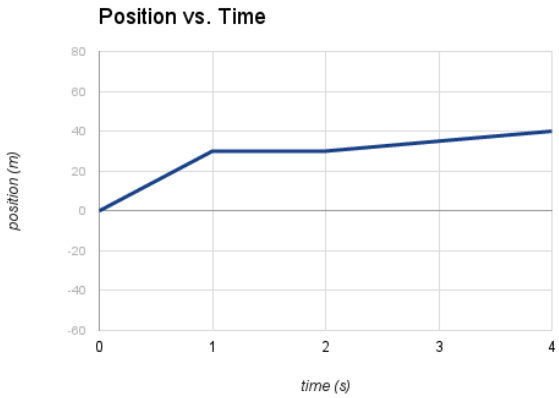
Ex. 1 Based on the Position vs. time graph, complete the v vs t graph

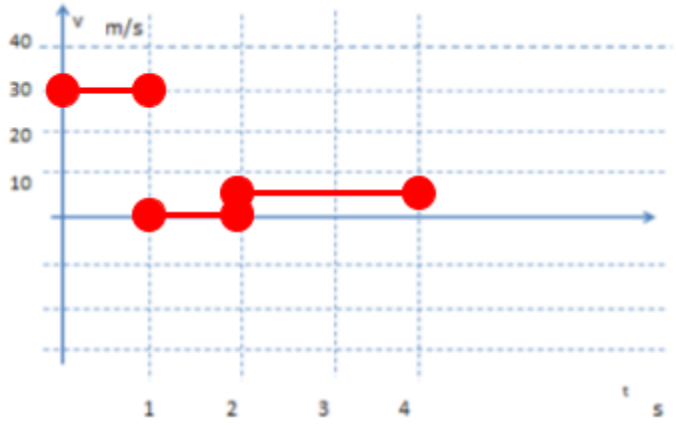


For the 0 - 2 s interval, the slope is 30 m/s. For the 2 - 4 s interval, the slope is 0 m/s.



Ex. 2 Complete the velocity versus time graph For the 0 - 1 s interval, the slope is 30 m/s. For the 1 - 2 s interval, the slope is 0 m/s. For the 2 - 4 s interval, the slope is 5 m/s ($\frac{40 - 30\ m}{4 - 2\ s}$).





Use these graphs for Ex. 3 on the next page.

Ex 3. What is the instantaneous velocity from the example above? Are there multiple ways to answer this question? It depends on which time interval you are using. There are many ways to answer this.

What is the average velocity from the example above? Are there multiple ways to answer this question?
Whole trip: 10 m/s; first interval: 30 m/s; second interval: 0 m/s; third interval: 5 m/s; **YES**

What needs to be specified to receive just one answer to the questions above?
You need to know the time interval in question.

Converting to position vs time graphs from velocity versus time

This section will be added once we have done it in class.

I. Do Now- What is an object's displacement when it travels at a velocity of 10 m/s for 5 seconds?

50 m in the positive direction (compared to the next question)

What is an object's displacement when it travels at a velocity of -10 m/s for 5 seconds?

50 m in the negative direction

II. Converting from position versus time to velocity versus time

a. What important steps needed to be taken to convert from a position versus time graph to a velocity versus time graph? Find the slope of each time interval; the slope is the average velocity for that interval.

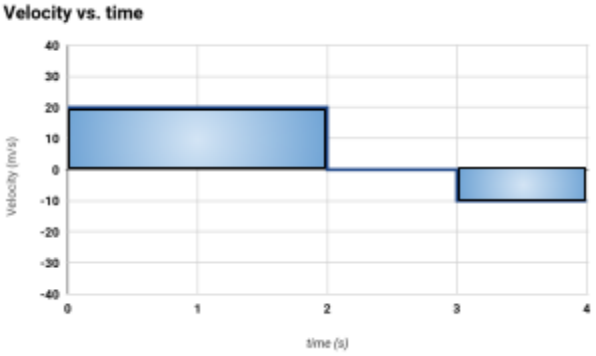
b. When an object was stopped, what did it look like on a position versus time graph? When an object was moving at constant velocity? It looks like a straight **horizontal** line; it looks like a straight line.

III. Converting from velocity versus time to position versus time

a. When an object's velocity is positive, its displacement is positive.

b. When an object's velocity is negative, its displacement is negative.

Ex. 1: Use the graph to answer the questions below.



1. From 0 to 2 seconds, how far did the object move?

40 m in the positive direction

2. From 2 to 3 seconds, how far did the object move?

0 m (it has no velocity)

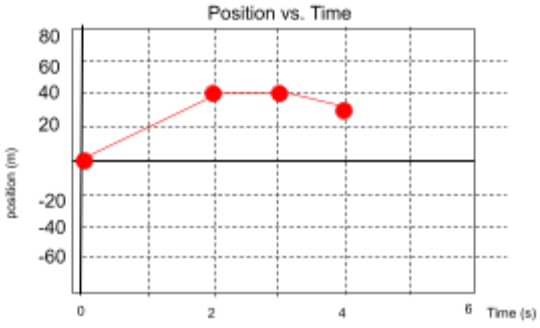
3. From 3 to 4 seconds, how far did the object move?

10 m in the negative direction

4. How does the graph tell us how far the object moved?

Look to the front board for shading instructions.

The area under each section of the velocity vs. time curve tells us the displacement for that section.

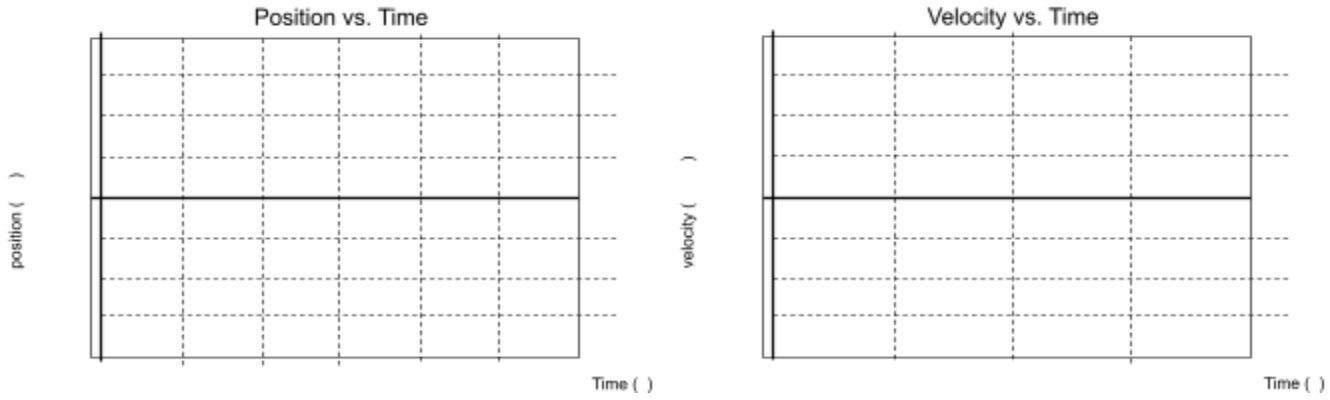


5. STOP! We will do the first graph together, then you will have time to practice.

Graph practice Round 1

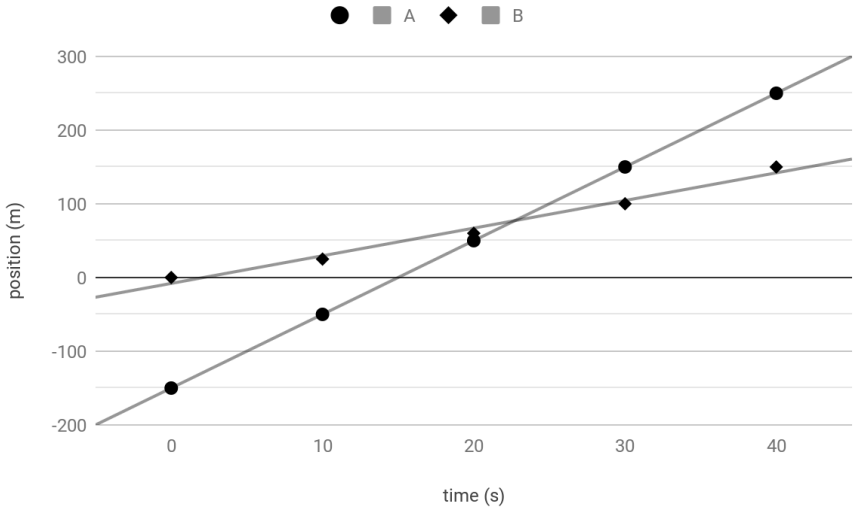
Rules: Draw a velocity versus time graph containing 2-4 horizontal segments and draw the answer on the inside of a folded piece of paper. You will hand off your v vs t graph to the next group, who will draw the two graphs below. When time is up,

hand the answer sheet to the next group so they can check their answers. Don't forget to **fully label your graphs.** We will settle all disputes as a class!



Round 2: Where do they meet?

Use the graph for the questions that follow.



Remember: the slope of a position v. time graph is the average velocity.

Remember: the equation of a straight line is y = b + mx

Reading the Graph

What is x_0 for A? -150 m

What is x_0 for B? About 0 m

Is A ahead or behind B when they start? A is behind

Which (A or B) is going faster? A is faster

When and where do they meet? (from the graph)

At ~22.5 seconds, ~75 meters

Translating to Math

Write an equation for the position of each:

A: $x_f = -150 + 10t$

B: $x_f = 0 + 3.75t$

Solve for where they meet:

Set the equations equal to each other. Solve for t, then plug into one of the two equations and solve for x_f. You should find that t = 24 s, and that x_f = 90 m.

D. Challenge Problems: Simultaneous Equations/System of Equations

Do Now: Think back to algebra. When you had two unknown variables, how many unique equations did you need to solve for those variables? If you had 5 unknown variables, how many equations?

Ex1. Two brothers are on a nature trail. Kevin is at the welcome center at the beginning of the nature trail and is walking at 2 miles per hour toward David, who is already at the 4.5 mile marker. David is jogging on the nature trail at a rate of 5 miles per hour toward the welcome center. Please answer the questions below.

a. Sketch the situation, including all the information from the prompt above.

b. Write an equation for the motion of Kevin.

$x_f = 0mi + (+\ 2\ mph)t$

c. Write an equation for the motion of David.

$x_f = 4.5mi + (-\ 5\ mph)t$

d. Use the equations from b and c to solve for the time it will be when the brothers meet each other.

2t = 4.5 - 5t ====> 7t = 4.5 =====> t = 0.6 hr

e. Using one of the equations from b or c, and your answer from d, where will the brothers meet?

$x_f = 0mi + (+\ 2\ mph)(0.6\ hr)$ = 1.2 mi (they meet at a position that is 1.2 miles on the positive side of the welcome center).

Ex2. A lion is chasing a gazelle. The lion has a top speed of 50 mph and the gazelle has a top speed of 30 mph. The gazelle was 0.5 miles away from the lion at the time the chase began, and the pair kept running until the gazelle was ultimately caught by the lion. Use the information to answer the questions below.

a. Sketch the situation, including all the information from the prompt above.

b. Write an equation for the motion of the lion.

$x_f = 0mi + (+\ 50\ mph)t$

c. Write an equation for the motion of the gazelle.

$x_f = 0.5mi + (+\ 30\ mph)t$

d. Use the equations from b and c to solve for the time it will be when the lion catches the gazelle.

50t = 0.5 + 30t ======> 20t = 0.5 ========> t = 0.03 hr

e. Using one of the equations from b or c, and your answer from d, where will the lion catch the gazelle?

$x_f = 0mi + (+\ 50\ mph)(0.03hr)$= 1.5 miles (they meet 1.5 miles on the positive side from where the lion started).

Closure: Why are these problems more difficult than the others we have done so far? Why are the answers to part b and c above essential to solving the problem?